

Modernizing Stacked Attention Networks for Visual Question Answering

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Introduction

This work is based on “*Stacked Attention Networks for Image Question Answering*” (Yang et al., 2016). Visual Question Answering (VQA) is a multimodal task that requires answering natural language questions based on image content. As described in the original paper, VQA differs from standard vision tasks in that different questions about the same image may require reasoning over different visual regions or their relationships.

As discussed in the original work, early VQA approaches relied on global image representations or single-step inference, which are insufficient for handling questions that require localized, question-dependent visual reasoning.

Building on this framework, it is important to note that **significant architectural advancements have been introduced** since the publication of the original SAN model. In this work, **we leverage more recent deep learning architectures and implementation practices** to enhance the original design, **resulting in improved performance over the baseline results reported in the original paper.**

Method and Implementation Overview

This work is based on the architecture proposed by **Yang et al** and was developed through multiple iterative versions. Each version introduced targeted refinements aimed at improving training stability, optimization behavior, and representational capacity, while preserving the core multi-step attention-based reasoning mechanism of the original model. The final system integrates these refinements into a coherent and robust implementation.

Implemented Architectures

We evaluate several architectures, reflecting both the original SAN design and successive extensions introduced during development:

1. Original SAN (Paper Results)

Reported results from the original *Stacked Attention Networks for Image Question Answering* paper, serving as an upper-bound reference for comparison.

2. SAN (Our Re-implementation)

A faithful PyTorch re-implementation of the original architecture using:

- ❖ A VGG16-based image encoder pretrained on ImageNet and kept frozen
- ❖ LSTM- or CNN-based question encoders
- ❖ One or two stacked additive attention layers
- ❖ Optimization with stochastic gradient descent (SGD) with momentum

This implementation follows the original design and training protocol as closely as possible, enabling direct comparison with the reported results.

3. ViT-SAN

A modernized variant of SAN in which the CNN-based image backbone is replaced with a frozen **Vision Transformer (ViT)**, while keeping:

- ❖ The original stacked attention mechanism
- ❖ The same question encoders
- ❖ An identical training and evaluation protocol

This variant isolates the effect of replacing convolutional visual features with transformer-based patch representations.

4. ViT-SAN-Q (Adam Optimizer)

An extended ViT-SAN variant that further introduces:

- ❖ Learnable query refinement across attention layers
- ❖ The Adam optimizer in place of SGD

This model is designed to improve convergence stability and enhance the effectiveness of multi-step reasoning, while retaining the overall SAN structure.

Datasets

All experiments were conducted using the **COCO-QA dataset**, built on **MS COCO 2014** images. No additional datasets were used, ensuring that observed performance differences across models reflect architectural and optimization changes rather than data variation.

Hyperparameters and Training Setup

Hyperparameters were initially selected to closely match those reported in the original SAN paper and were subsequently refined during development. The dimensionality of both visual and textual embeddings was fixed at **$d = 500$** . Question encoding was performed using either:

- ❖ a single-layer unidirectional LSTM with hidden size 500, or
- ❖ a CNN-based encoder with multiple temporal convolution filters.

Unless stated otherwise, models used **two stacked attention layers**, as deeper stacks did not yield additional gains.

For SAN and ViT-SAN baselines, optimization was performed using **SGD with momentum 0.9** and **weight decay 5×10^{-4}** . In later development stages, several engineering-level improvements were introduced, including:

- ❖ mixed-precision training (AMP) to optimize memory and speed.
- ❖ Gradient norm clipping (threshold 5.0) to stabilize the training of the LSTM/CNN encoders, preventing gradient explosions.
- ❖ An improved batching and padding pipeline.

These changes improved training stability and computational efficiency without altering the underlying model formulation.

The ViT-SAN-Q variant replaces SGD with **Adam**, aiming to improve optimization dynamics and convergence behavior.

Evaluation Metrics

Evaluation was primarily performed using **classification accuracy**, measuring the proportion of exactly correct predictions over the fixed answer vocabulary. This follows the standard VQA formulation as a multi-class classification problem with single-word answers and is consistent with the original SAN evaluation protocol.

In addition, we report **Wu-Palmer Similarity (WUPS)** scores at two thresholds:

- ❖ **WUPS@0.9** (strict semantic similarity)
- ❖ **WUPS@0.0** (relaxed semantic similarity)

WUPS is computed using WordNet-based semantic similarity and accounts for cases where predicted answers are semantically related but not lexically identical.

During training, **cross-entropy loss** was monitored on both training and validation sets to assess convergence and overfitting behavior. Final model comparison was based on accuracy and WUPS scores evaluated on the held-out test set, following the reporting conventions of the original paper.

Results and Analysis

Best Results by different evaluation metrics:

Accuracy (ACC)

Model Family	LSTM (1)	CNN (1)	LSTM (2)	CNN (2)
SAN (Original Paper)	59.6	60.7	61.0	61.6
SAN (Our Implementation)	56.35	55.60	56.60	57.32
ViT-SAN	61.45	61.10	60.90	61.91
ViT-SAN-Q (Adam)	61.29	63.06	61.98	62.44

WUPS 0.9

Model Family	LSTM (1)	CNN (1)	LSTM (2)	CNN (2)
SAN (Original Paper)	69.6	70.6	71.0	71.6
SAN (Our Implementation)	63.75	63.87	63.79	65.01
ViT-SAN	68.83	68.58	68.87	69.68
ViT-SAN-Q (Adam)	68.91	70.51	69.93	70.22

WUPS 0.0

Model Family	LSTM (1)	CNN (1)	LSTM (2)	CNN (2)
SAN (Original Paper)	90.1	90.5	90.7	90.9
SAN (Our Implementation)	88.97	89.26	89.15	89.35
ViT-SAN	90.56	90.48	90.54	90.78
ViT-SAN-Q (Adam)	90.50	91.06	90.78	90.91

Key Observations and Analysis

The normalized comparison across all model families reveals several consistent and data-driven trends that clarify how architectural choices influence performance in Visual Question Answering (VQA). The analysis below directly reflects the final results reported for Accuracy, WUPS@0.9, and WUPS@0.0.

Effect of Stacked Attention

Increasing the number of attention layers from one to two generally leads to performance improvements, though the effect is moderate rather than uniform. In the original SAN paper, two-layer models consistently outperform their single-layer counterparts across all metrics, confirming the benefit of multi-step attention. In our re-implementation, the improvement from stacking is smaller but still observable, particularly for CNN-based question encoders. For ViT-based models, stacked attention continues to provide gains, most clearly reflected in Accuracy and WUPS@0.9, supporting the role of iterative attention refinement in visual reasoning.

Question Encoder Comparison

CNN-based question encoders tend to outperform LSTM-based encoders across most configurations, with the effect becoming more pronounced in stronger visual backbones. While the difference between CNN and LSTM encoders is modest in the original SAN and our re-implementation, CNN encoders consistently achieve higher Accuracy and WUPS scores in ViT-SAN and ViT-SAN-Q. This suggests that CNN question encoders may better complement transformer-based visual features by capturing localized semantic patterns that align effectively with globally contextualized image representations.

Impact of Visual Backbone Modernization

Replacing the VGG-based image encoder with a **Vision Transformer (ViT)** yields a clear and consistent performance improvement across all three evaluation metrics. ViT-SAN models outperform their SAN counterparts by approximately 4-5% in Accuracy, with corresponding gains in WUPS@0.9 and WUPS@0.0. These results highlight the importance of global visual representations in VQA tasks, as ViT models capture long-range spatial relationships early in the pipeline, providing more informative features for the stacked attention mechanism.

Query Refinement and Optimization Strategy

The ViT-SAN-Q variant introduces learnable query refinement across attention layers and replaces SGD with the Adam optimizer. This combination results in the strongest overall performance among all evaluated models. The largest gains are observed in Accuracy, particularly for the CNN-based configurations, while consistent improvements are also reflected in semantic similarity metrics. These findings indicate that adaptive query updates enable more precise alignment between question semantics and visual evidence, and that Adam optimization improves convergence stability when additional learnable components are introduced.

Consistency Across Evaluation Metrics

Crucially, performance improvements are not confined to exact-match accuracy. Gains are consistently reflected in both WUPS@0.9 and WUPS@0.0, demonstrating that the observed improvements correspond to better semantic grounding and reasoning rather than superficial answer matching. This consistency across strict and relaxed evaluation metrics reinforces the conclusion that the architectural enhancements lead to genuine improvements in visual-language understanding.

Conclusion

This project demonstrates that the core principles of the original **Stacked Attention Network (SAN)** remain highly effective, while also benefiting significantly from modern architectural and optimization advances. Through systematic experimentation and normalized comparison, we show that performance gains arise from the **synergistic effect of improved visual representations, multi-step attention reasoning, and adaptive optimization**.

A key outcome of this work is the **successful re-implementation of the original SAN architecture in PyTorch**. Our implementation faithfully reproduces the qualitative and quantitative trends reported in the original paper, including the benefits of stacked attention layers and the relative behavior of LSTM- and CNN-based question encoders. Although minor performance gaps exist compared to the reported results—expected due to differences in training dynamics and implementation details—the reproduced results closely align with the original findings, validating the correctness and robustness of our implementation. Building upon this solid baseline, we explored principled extensions to the original model. Among all evaluated variants, **ViT-SAN-Q with a CNN-based question encoder and two stacked attention layers** achieves the best overall performance across all evaluation metrics. These findings confirm that attention-based reasoning remains a powerful paradigm for VQA, and that integrating transformer-based vision models together with learnable query refinement significantly enhances visual-language understanding.

Overall, the results validate both the original SAN design and the proposed improvements, offering a clear and principled pathway toward more **robust, interpretable, and scalable visual-language reasoning systems**. The successful re-implementation and extension of SAN further demonstrate the model's flexibility and continued relevance in the context of modern deep learning architectures.

Challenges Faced

- ❖ **Long training time per experiment** - Training both SAN and ViT-based variants required significant computational time, especially due to large input resolutions and multi-step attention, which limited the number of experiments that could be conducted.
- ❖ **Slow iteration cycle for architectural changes** - Even minor architectural or optimization modifications necessitated full retraining to reliably assess their impact, resulting in a slow and costly development cycle.
- ❖ **Complex dataset structure and vocabulary construction** - The COCO-QA dataset required careful handling of image-question-answer mappings, vocabulary construction, and padding strategies, increasing implementation complexity and the potential for subtle preprocessing errors.

What we learned

- ❖ Re-implementing and extending the Stacked Attention Network provided practical insight into how architectural design, optimization, and dataset characteristics interact in Visual Question Answering systems.
- ❖ One key lesson was that **training stability and generalization are as important as architectural improvements**. Stronger visual backbones, particularly ViT-based models, converged faster and achieved lower training loss, but also exhibited a higher tendency to overfit on COCO-QA. This required careful monitoring of validation performance and the use of stabilization techniques to ensure that improvements reflected genuine reasoning gains rather than memorization.
- ❖ We observed that **overfitting was driven primarily by the visual backbone and optimization strategy**, rather than by the stacked attention mechanism itself. Increasing the number of attention layers did not significantly worsen generalization, suggesting that stacked attention functions as a structured reasoning process rather than uncontrolled capacity growth.
- ❖ Another important takeaway was the role of implementation-level decisions. Techniques such as mixed-precision training, gradient norm clipping, and improved batching were essential for reliable experimentation and fair model comparison. Without these measures, training behavior was unstable and performance differences across models were difficult to interpret.
- ❖ Finally, using both **accuracy and semantic similarity metrics (WUPS)** proved critical for understanding model behavior. Improvements that were consistent across strict and relaxed WUPS thresholds indicated better semantic grounding, while divergence between training accuracy and WUPS helped identify early signs of overfitting.
- ❖ Overall, the project reinforced that modernizing a classic architecture requires not only improved components, but also careful attention to optimization, evaluation, and generalization behavior

suggestions for further work

1. End-to-End Fine-Tuning of the Visual Backbone

While the ViT backbone was kept frozen for fair comparison, fine-tuning selected transformer layers could further improve alignment between visual representations and question-guided attention.

2. Evaluation on More Challenging VQA Benchmarks

Extending evaluation to datasets such as VQA v2 or GQA would test the scalability and reasoning robustness of SAN-style architectures in more complex settings.